University of St.Gallen

Using Computer Simulation

for Understanding Organizations as Complex Systems

Week 6: How to run simulation experiments and analyse simulation data?

Stephan Aier

From insight to impact.

Structure

Date	Topic	Deliverables	Literature	Responsible
19.02.	Introduction to the course, theory foundations, simulation demos	-	papers 1, 2	Stephan & Marcel
26.02.	How to apply agent-based modelling?	-	book chapter 1	Marcel
05.03.	How to implement simulation models with NetLogo?	-	book chapter 2 (3, 4, 5)	Marcel
12.03.	Independent case work (*no input session*)	-		-
19.03.	Intermediate group presentation and feedback on case/model	Presentation		Stephan & Marcel
26.03.	**How to run simulation experiments and analyse simulation data?**	-	book chapter 6	Stephan
	(*break*)			
16.04.	Intermediate group presentation on implemented model and analysis	Presentation		Stephan & Marcel
23.04.	How to verify and validate simulation models?	-	book chapter 7	Stephan
30.04.	Independent case work (*no input session*)	-		-
07.05.	Final case presentation in groups and reflection of learnings	Presentation, Model, Analysis		Stephan & Marcel
14.05.	Ascension Day (*no input session*)	-		-
21.05.	Oral exams	Oral Exam		Stephan & Marcel

Paper 1: Beese, J., Haki, M. K., Aier, S., & Winter, R. (2019). Simulation-Based Research in Information Systems: Epistemic Implications and a Review of the Status Quo. Business & Information Systems Engineering, 61(4), 503-521.
Paper 2: Haki, K., Beese, J., Aier, S., & Winter, R. (2020). The Evolution of Information Systems Architecture: An Agent-Based Simulation Model. MIS Quarterly, 44(1), 155-184.
Book: Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press.

Agenda

1	Recap of Weeks 3/5	4
2	Planning and Running Experiments in NetLogo	9
3	Analysis of Simulation Data	15
4	Getting our hands dirty	18

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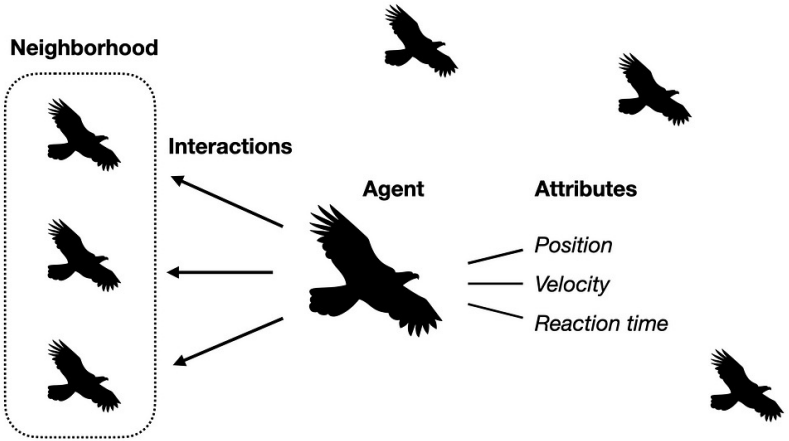
Before we start to code…

Remember: While modelling, we **must make pragmatic choices**. The selection of attributes is never complete, but it should be well-reasoned.

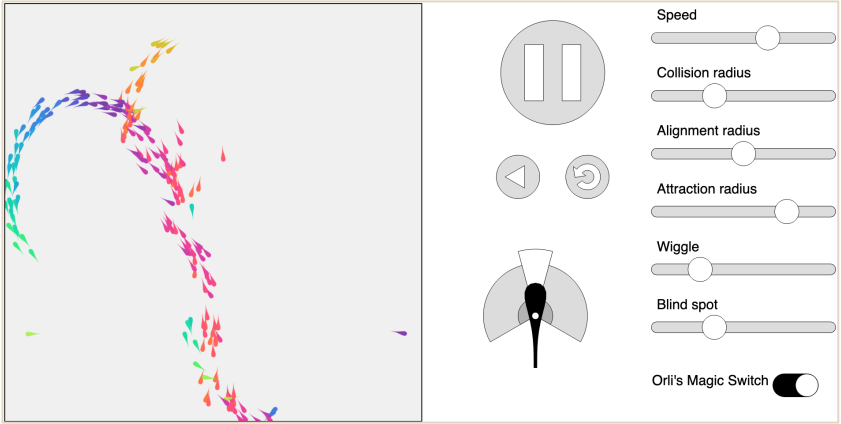
Start with the simplest version of your model that you can come up with (very few attributes and agent types).

Increase the level of specificity of your conceptualisation of the model to such a degree that you can 'run' the simulation on paper, including attribute values and the respective update mechanisms.

Only then, you start to code…



Source: towarddatascience.com



Source: complexity-explorables.org

Recap week 3

Developing the right intuitions for NetLogo: Overview

3-Tab structure:

- Interface (visualisation, interaction)
- Info (description of the ABM)
- Code (logic of the ABM)

Wolf Sheep Predation (Docked Hybrid) - NetLogo

File Edit Tools Zoom Tabs Help

Interface Info Code

Find... Check Procedures Indent automatically Code Tab in separate window

```
;; Sheep and wolves are both breeds of turtle.
breed [sheep a-sheep] ;; sheep is its own plural: we use "a-sheep" as the singular.
breed [wolves wolf]
turtles-own [energy]  ;; both wolves and sheep have energy

to setup
  clear-all
  ask patches [ set pcolor green ]
  set-default-shape sheep "sheep"
  create-sheep initial-number-sheep [ ;; create the sheep
    ;; then initialize their variables
    set color white
    set size 1.5  ;; easier to see
    set label-color blue - 2
    set energy 1 + random sheep-max-initial-energy
    setxy random-xcor random-ycor
  ]
  set-default-shape wolves "wolf"
  create-wolves initial-number-wolves [ ;; create the wolves
    ;; then initialize their variables
    set color black
    set size 1.5  ;; easier to see
    set energy random (2 * wolf-gain-from-food)
    setxy random-xcor random-ycor
  ]
  display-labels
  reset-ticks
end

to go
  if not any? turtles [ stop ]
  ask sheep [
    move
    death
    reproduce-sheep
  ]
  ask wolves [
    move
    set energy energy - 1  ;; wolves lose energy as they move
    catch-sheep
    death
    reproduce-wolves
  ]
```

Some observations from group presentations

1. Make sure you can list and specify your model's **basic components** (agents, environment, interaction behavior).
2. What is the **"source of emergence"** in your model? Usually this means, the behavior of one agent depends on the behavior of another agent, which again influences the behavior of the first agent (in the next tick).
3. Start small: What is really the **minimal model** that you can start implementing? It will get complicated on its own. Having a minimal but running model allows you to create data and "improve" your model based on that data.
4. Describe your model using **Pseudo Code** to describe what should happen in **one tick**. This is basically the go procedure.
5. Think about your **manipulations**. In other words, what are your independent variables that you want to vary in certain ranges?
6. Think about your **outcomes/goals/effects**. In other words, what are your dependent variables that you consider the effects of your manipulations? Note: you often want more than one since the world is full of trade-offs.
7. Is the set of independent and dependent variables suited to answer your **research questions**?
8. Very often you do not necessarily want to "find the optimal value for *x to maximize y*". More often it is about **understanding the underlying mechanisms** how *x is related to y, in the context of z, … .*

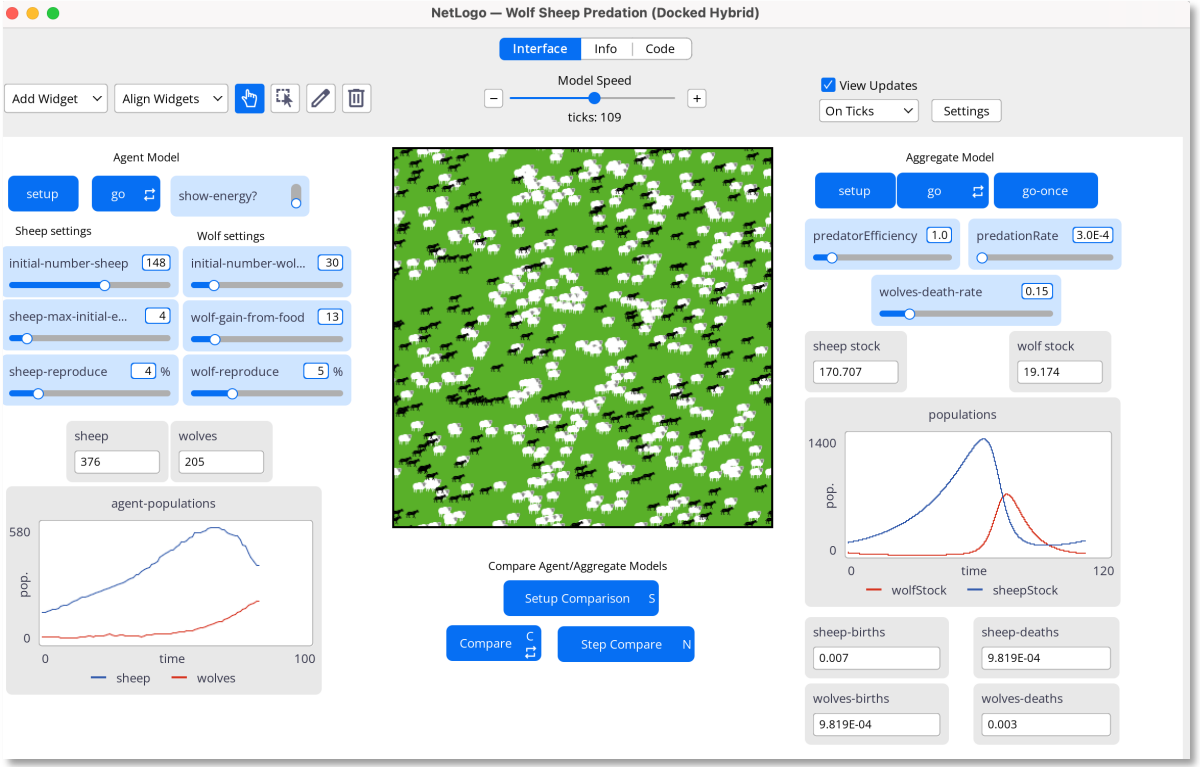
Developing the right intuitions for NetLogo: The Interface

3-Tab structure:

- **Interface** (visualisation, interaction)
- Info (description of the ABM)
- Code (logic of the ABM)

The interface comes with:

- Buttons (e.g., setup, go)
- Sliders (e.g., #agents, or speed)
- Monitors/Plots (e.g., tracks #agents)
- A visual representation of the state of the simulation (environment)



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Where are we now?

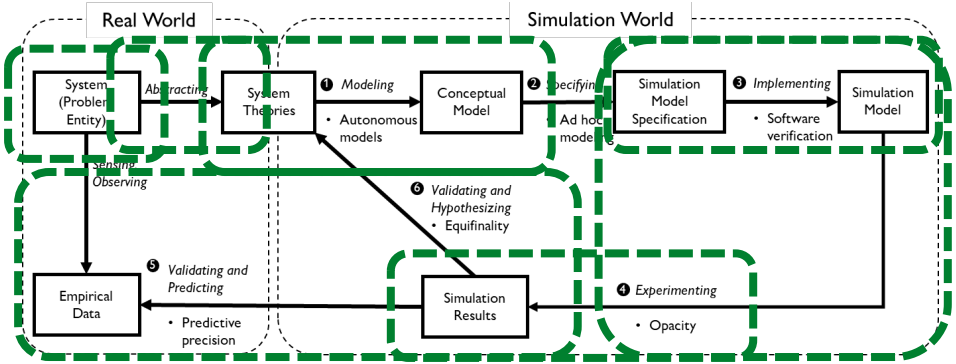
Approach

1. Identify a phenomenon and formulate a question

2. Narrow down the scope of the phenomenon

3. Identify and define actors (incl. relevant attributes and behaviours), environment (incl. non-spatial vs. spatial, dynamic vs. constant), & interactions (including stochasticity in actor, env., interactions).

4. Implement the model (NetLogo, week 3)
5. Calibrate the model (week 4)
6. Run simulations (week 6)
7. Validate the model & draw conclusions (week 8)



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What is an experiment?

Experiment

- Controlled environment
- We manipulate ONE variable and try to see that changing this variable's value has no impact on the outcome of the experiment (*null hypothesis*)
- If the outcome of the experiment changes, it must have been because of that ONE variable

Advantages

- Repeated testing of cause-effect-relations/theories in a controlled environment
- Observation of behavior in hypothetical situations

Disadvantages

- Deep external validity (laboratory experiment) or deep internal validity (field experiment)
- Often limited to single/very few variables

What are the goals of experiments in agent-based simulations?

1. We want to **remove many of the limitations** of traditional experiments (low number of variables)

2. We do not only want to see whether there is an effect, but we also want to **uncover the potentially complex underlying mechanisms** that lead to the observed effects.

3. Therefore, we also control for many variables, and we also create **experimental settings that would be impossible in the real world**.

4. We may even **predict outcomes** (requires calibration with empirical data).

What are the implications for the number of experiment settings?

1. We often prefer (quasi) **continuous variables** over discrete (scenarios) variables because this offers more (interesting) data analyses.

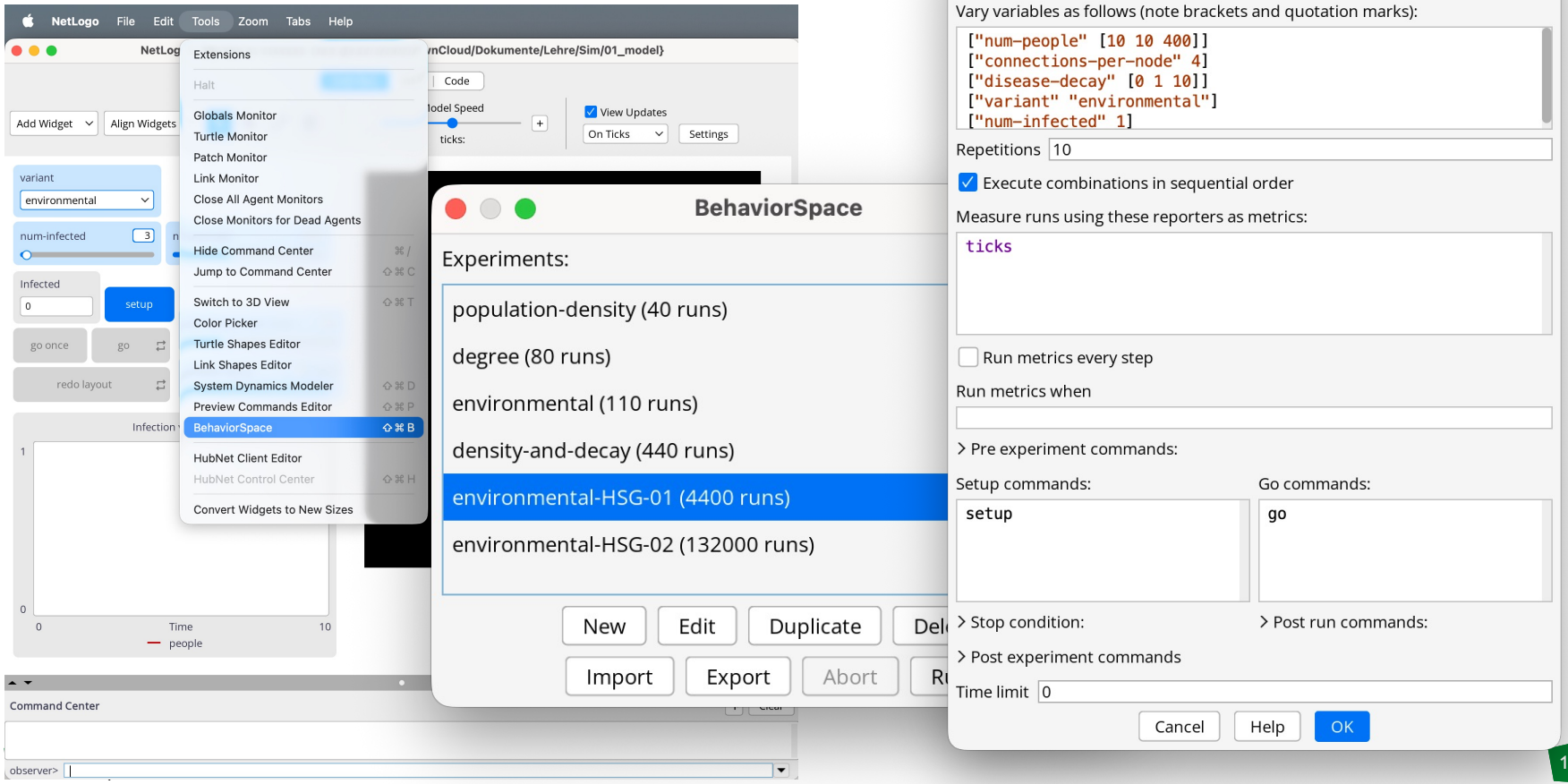
2. We create the **combinations of all possible values** of the independent variables:
   1. Var1 = 2, 4, 6, 8, 10
   2. Var2 = 10, 50, 100
   3. Experimental settings = (2, 10), (4, 10), (6, 10), (8, 10), (10, 10), (2, 50) … (10, 100)

   This may rather quickly lead to a very high number of experimental settings (runs in NetLogo) – this may easily become impossible to reasonably run on most machines → **limit variables, ranges, or resolution**

3. If our model contains **random values**, we usually want to **run every combination** of possible values **several times**.

The Behavior Space

Setting up experiments in NetLogo



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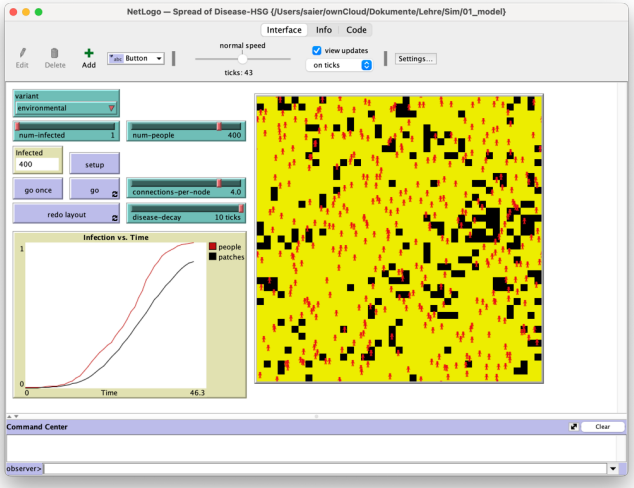
Data analysis within NetLogo or elsewhere?

Development / Testing

Using the **build-in** diagrams and slider in **NetLogo** is fast and convenient.

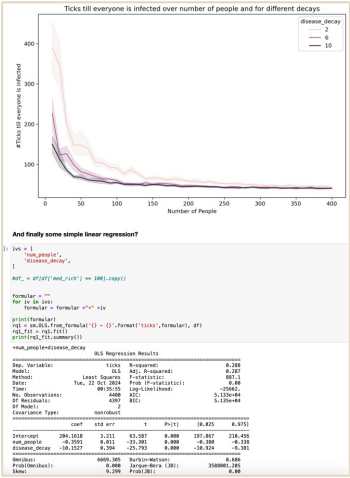


Analyzing Data of “final model”

We usually want some more powerful means of data exploration, visualization, and analysis.

This may be **Excel** (but this is rather inconvenient for repeated data generation)

Or any statistics package such as R, or here mostly some kind of **Python libraries (in Jupyter Notebooks)**



A simple Jupyter Notebook for data analysis to start with

Let's explore our little Demo Notebook.

Some simple analyses of simulation data

This notebook will provide some starting points for data visualization and regression analysis of simulation data. It is by no means a statistics course. There are of course many more ways (some of those may be more appropriate for **your dataset**) than shown here.

It often is a good starting point to consult the documentation of the functions used here to see what else they can do or how they can be adapted to our data/needs.

Setting up our notebook

Prepare everything including our raw data.

First some packages to import

In [1]:
```
# some basic stats/datascience
import numpy as np
import pandas as pd
import statsmodels.api as sm

# and some data visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

Then we need to load our simulation data

Loading csv files into a dataframe is rather straight forward.

In [2]:
```
# the filename is relative to the notebook
# hardcoded in this case
filename = '../02_data/Spread of Disease-HSG environmental-HSG-01-table.csv'

# read the csv file into a dataframe and skip the first rows
df = pd.read_csv(filename, skiprows=6)

# we want see all columns in our DataFrame later on
pd.set_option("display.max_columns", None)
```

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Assignment

1. For your group project's model, identify variables
    1. The independent variables (discrete, continuous, etc.)
    2. The dependent variables (every step, after *x* steps, min, max, avg, last, etc.)
    3. Will those be sufficient to answer your (research) questions?

2. Set up the behaviour space of your model
    1. What are the min/stepsize/max values in your experiment
    2. How many runs does this result in?
    3. Do you have random values in your model? → do you need to run through the behaviour space more than once?

3. Data Analysis
    1. What kind of data visualizations do you plan for?
    2. What kind of descriptive, statistical analyses do you plan for?

What's next?
Week 7: Group Presentations, part II

Week 7 (after break): Group Presentations

- A **short** recap of your idea/model
  - agents, environment, interaction behavior;
  - research questions;
  - variables to manipulate/independent variables, variables to measure/dependent variables
- A short demo of the current state of your model in NetLogo
  - what works, what not, what is still missing;
  - can you successfully manipulate independent variables, are the resulting values of dependent variables reasonable, can you explain the changes?
- Your next steps

Backup: Agenda